ダス・ソハン ／ SOHAN DAS

ロボティクスエンジニア *Robotics Engineer*

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(就労ビザ取得可能)

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■ 履歴書基本情報 ／ Personal Information

氏名 / Full Name	Sohan Das (ダス・ソハン)	**国籍 / Nationality**	Indian (インド国籍)
現住所 / Address	Manchester, UK	**在留資格 / Visa Status**	Student Visa → Tech Visa 取得予定
日本語能力 / Japanese	基礎レベル (JLPT N5-N4 相当)	**英語能力 / English**	ビジネスレベル (Native-equivalent)
最終学歴 / Education	MSc Robotics, Univ. of Manchester (2025–2026)	**就業可能時期 / Available**	2026年9月以降 / September 2026

■ 職務要約 ／ Professional Summary

マンチェスター大学ロボティクス修士課程に在籍するロボティクスエンジニアです。制御システム・センサーフュージョン・組み込みエレクトロニクスを専門とし、ROS2・MATLAB・Python・PCB設計の実務経験を有します。自律走行ロボット・UAV制御・ヒューマンロボットインタラクション（HRI）の研究開発に携わった実績があり、日本のテクノロジー企業において即戦力として貢献できます。

MSc Robotics student at the University of Manchester, specialising in control systems, sensor fusion, and embedded electronics. Experienced in ROS2, MATLAB/Simulink, Python, and custom PCB design. Track record in autonomous navigation, UAV position control, and Human-Robot Interaction (HRI) research. Eager to contribute to Japan's cutting-edge robotics and tech sector.

■ 学歴 ／ Education

Master of Science in Robotics | University of Manchester, United Kingdom
September 2025 – September 2026

- Coursework: Advanced Control Systems, Sensor Fusion, Cognitive Robotics, Computer Vision, Embedded Systems
- Research focus: Human-Robot Interaction (HRI) and autonomous robotic navigation

Bachelor of Technology — Electrical & Electronics Engineering | VIT University, India
August 2020 – July 2024 | GPA: 8.46 / 10.0 (Top Quartile)

- Specialisation in embedded systems, robotics, power electronics, and control engineering
- Final-year capstone thesis: Deep Reinforcement Learning (DDPG, SAC, TD3) for vibration suppression in rotary flexible link systems using Quanser SRV02 physical hardware — supervised by Dr. Vinodh Kumar E
- Independent research (separate from thesis): Arm Motion Planning of Humanoid Robot — resulted in peer-reviewed publication (DOI: 10.13140/RG.2.2.33678.54082)

- Relevant modules: Digital Signal Processing, Microcontroller Programming, Power Systems, Robot Kinematics

■ 職務経歴 ／ Professional Experience

Virtual Robotics & Control Engineer | Forage (Remote Simulation Programme)
February 2026 – March 2026

- Completed industry-standard engineering workflows simulating real-world robotics control scenarios
- Implemented embedded control solutions applying structured problem-solving and iterative debugging
- Optimised software modules to meet industry-level performance and code-quality benchmarks
- Demonstrated proficiency in translating engineering requirements into executable, maintainable code

Mathematics & Computer Science Teaching Volunteer | International Association of Lions Club
July 2024 – December 2024

- Delivered structured STEM lessons to students, adapting technical concepts for varied learning abilities
- Developed and iterated lesson materials focusing on logical thinking and computational problem-solving
- Supported underserved communities in gaining foundational computer science and mathematics literacy

RPA Developer Intern | Automation Anywhere
June 2022 – July 2022

- Developed robotic process automation (RPA) workflows to streamline repetitive enterprise tasks
- Gained practical experience with the Automation Anywhere platform and bot deployment lifecycle
- Collaborated with engineering teams to identify automation opportunities and measure efficiency gains

■ 主要プロジェクト ／ Key Technical Projects

Leo-Rover Robot	Hardware integration and software control of a modular rover platform using ROS2 and Python
BTech Thesis: DRL Vibration Control	Comparative analysis of DDPG, SAC and TD3 deep RL algorithms for vibration suppression on Quanser SRV02 physical rotary flexible link hardware — MATLAB/Simulink + RL Toolbox; SAC achieved superior sample efficiency and robustness
Autonomous Navigation (ROS2 Nav2)	End-to-end autonomous navigation pipeline using ROS2 Nav2 stack — mapping, localisation (SLAM), and path execution
UAV Position Controller	Cascade PID controller with Disturbance Observer for real-time aerial object stabilisation and trajectory tracking
Path Planning (A* Algorithm)	Implemented A* search algorithm for optimal robot path planning in grid-based and dynamic environments
EV Battery Management System	Designed a Battery Management System (BMS) for electric vehicles including cell balancing and safety monitoring
Floor Cleaning Robot	Built an autonomous floor-cleaning robot with obstacle avoidance sensors and coverage path planning
Humanoid Arm Motion Planning	Research project and published paper on motion planning for a 7-DOF humanoid robot arm (DOI referenced above)

■ 技術スキル ／ Technical Skills

プログラミング言語 / **Programming Languages:**

Python	MATLAB	Simulink
Java	ROS2	R
C++	Nav2	SLAM
Embedded C	Control System / PID	Kinematics
SQL	HTML / CSS / JS	

ハードウェア・ツール / **Hardware & Tools:**

Arduino	Raspberry Pi	Sim2Real
STM32 Microcontroller	Real-time Control	High-DOF Robots
Sensor Fusion	System Integration	Computer Vision

ソフトウェア・その他 / **Software & Other:**

Automation Anywhere	PCB Design (KiCad)	OrCAD PSpice
Gazebo / RViz	FANUC	Cyberbotics
MuJoCo	RoboDK	Linux (Ubuntu)
SLAM	Nav2 Stack	Git / GitHub

■ 言語能力 ／ Language Skills

GB 英語 / **English** Business Proficient (C1+)	BD ベンガル語 / **Bengali** Native (母国語)	IN ヒンディー語 / **Hindi** Fluent (流暢)	JP 日本語 / **Japanese** Basic / 基礎 (学習中)

■ 研究・出版物 ／ Research & Publication

Arm Motion Planning of Humanoid Robot | Peer-Reviewed Publication

DOI: 10.13140/RG.2.2.33678.54082 | 2024

- Conducted original research on motion planning algorithms for a 7-DOF humanoid robot arm
- Developed and validated trajectory generation methods for collision-free motion in constrained workspaces
- Findings applicable to industrial robotics, collaborative robots (cobots), and HRI research domains

卒業論文 / BTech Thesis | Comparative Analysis of Deep Reinforcement Learning Algorithms for Vibration Suppression in Rotary Flexible Link Systems

VIT University | Supervisor: Dr. Vinodh Kumar E | April 2024

- Implemented and comparatively evaluated three deep RL algorithms — DDPG, SAC, and TD3 — for active vibration suppression on a physical Quanser SRV02 rotary servo system with FlexGage module
- Developed complete MATLAB/Simulink simulation environment using Reinforcement Learning Toolbox and Deep Learning Toolbox; trained actor-critic neural network policies for continuous action-space control
- SAC achieved highest sample efficiency and disturbance robustness; TD3 demonstrated fastest convergence — findings provide a benchmark framework for DRL-based robotic control system design
- Directly relevant to AIRoA: DRL policy training on physical robotic hardware, real-time control loops, and ML model evaluation on continuous-action robotic systems

■ 志望動機・自己PR ／ Motivation & Self-Introduction

日本語での自己PR：
私はロボティクスと制御工学に強い情熱を持ち、日本の先進的なテクノロジー企業で経験を積みたいと考えています。ROS2・Python・組み込みシステムの実務経験を活かし、貴社の製品開発・研究開発に貢献できると確信しています。日本語を積極的に学習中であり、グローバルかつ多文化なチームでの協働にも慣れています。

Self-PR (English):
I am a passionate robotics engineer who combines strong academic foundations with practical engineering experience. My work spans autonomous navigation, UAV control, and human-robot interaction — areas in which Japanese technology companies are global leaders. I am self-driven, detail-oriented, and committed to continuous learning, including actively studying Japanese. I thrive in collaborative, cross-cultural environments and am excited to bring my skills to Japan's innovative tech ecosystem.

■ 資格・課外活動 ／ Certifications & Activities

Automation Anywhere — RPA Developer	Industry internship certification in robotic process automation (2022)
Forage — Engineering Simulation	Virtual industry experience in robotics and control engineering workflows (2026)
ROS2 — Online Certificate	Certified course in Robot Operating System 2 (ROS2) — robotics middleware, publishers, subscribers, navigation
Embedded C Programming — Online Certificate	Formal certification in Embedded C covering memory management, peripherals, interrupts, and RTOS concepts
Robot Motion Planning & Control — Online Certificate	Certified course covering trajectory planning, kinematics, dynamics, and motion control algorithms for robotic systems
Supervised Machine Learning — Online Certificate	Certification in supervised ML methods including regression, classification, and model evaluation techniques
Robotics with Arduino — Online Certificate	Hands-on certification in Arduino-based robotics: sensors, actuators, motor drivers, and real-time embedded control
Wix Website Builder — Online Certificate	Certified in professional web development and deployment using the Wix platform
CPD Portfolio (Active)	Continuous Professional Development portfolio — all certifications, courses, and skills verified and logged (see CPD link at top)
Japanese Language Study	Actively studying Japanese (Hiragana, Katakana, basic conversation) — targeting JLPT N4.

本履歴書の記載内容はすべて事実です。 All information stated herein is accurate and truthful.

Sohan Das 2026年4月 April 2026